Lab 1 - Extrude - Personalised Keychain Token

Course: Innovating Product Development with 3D Printing and Additive Manufacturing (TGS-2022017597) Topic 01: Overview of Additive Manufacturing and 3D Scanning TSC objective: A1 - determine material and design considerations while modelling a printable part with the extrude technique

Goal

Model an elliptical keychain token, add your initial, then slice and print it.

What you'll build

A personalised elliptical keychain token, sliced and 3D printed in PLA.

Tools: Fusion 360, Ultimaker Cura, FDM printer (PLA)

Steps

1. Create a new design in Fusion 360 and start a sketch on the XY plane.

```
Fusion 360 > File > New Design > Create Sketch > XY plane
```

2. Draw an ellipse about the origin - approximately 50 mm long and 30 mm wide.

```
Sketch > Ellipse - major 50 mm, minor 30 mm
```

3. Add a 4 mm circle near one end of the ellipse for the keyring hole, then finish the sketch.

```
Sketch > Circle - Ø 4 mm > Finish Sketch
```

4. Extrude the ellipse profile (excluding the circle) to 3 mm to form the token body.

```
Create > Extrude - profile: ellipse, distance 3 mm
```

5. Sketch the first letter of your name on the token's top face and extrude it 1 mm.

```
Create Sketch > Text - height 18 mm > Extrude 1 mm (Join)
```

6. Export the body as an STL mesh for slicing.

```
File > Export > STL  (or Utilities > Make > 3D Print)
```

7. Open the STL in Cura, keep the flat face on the build plate, and slice with the default 0.2 mm profile.

```
Cura > Open File > token.stl > Slice
```

8. Save the G-code and print the token; remove it from the bed once cooled.

```
Cura > Save to Disk > print
```

Evidence to capture

- Fusion 360 screenshot showing the ellipse, 4 mm hole and raised initial.

- Cura Preview screenshot confirming flat, support-free orientation.
- Photo or note confirming the raised initial and clear keyring hole.

Test it

The sliced preview shows the token printing flat with no supports; the printed token carries your raised letter and the keyring hole is clear.

Files in this folder

- token.stl - Original sample solution generated by the build pipeline (672 triangles).